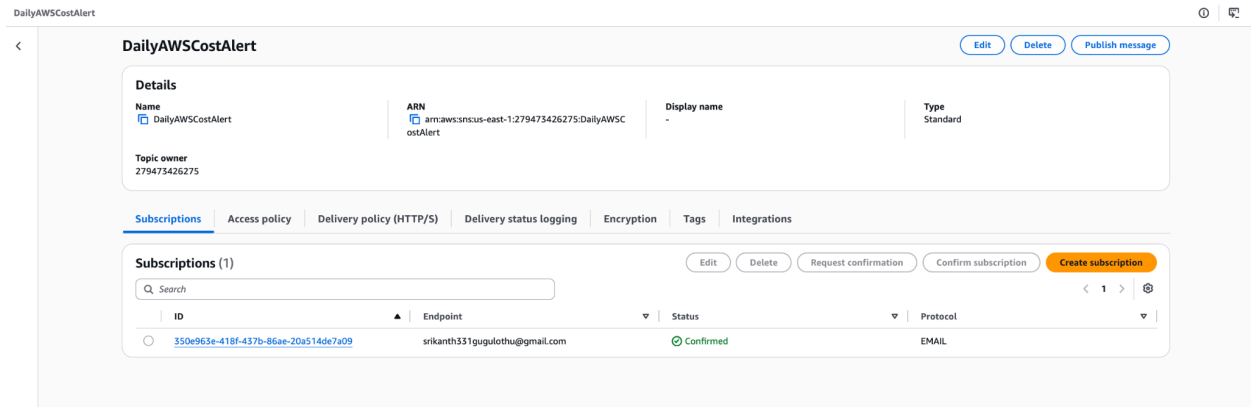


## Task 5: Daily AWS Cost Alert Using Cost Explorer API and SNS

### Step 1: Create SNS Topic

Created an SNS topic named **DailyAWSCostAlert** and subscribed an email address using the Email protocol



### Step 2: Create IAM Policy

Create an IAM role with the following configuration:

- **Trusted Entity:** AWS Service
- **Use Case:** Lambda
- **Permissions Policy:** **AWSLambdaBasicExecutionRole**
- **Role Name:** **S3BucketCleanupRole**

nupRole 0

### S3BucketCleanupRole [Info](#)

Allows Lambda functions to call AWS services on your behalf.

[Delete](#)

#### Summary

**Creation date**  
July 23, 2026, 12:49 (UTC+05:30)

**Last activity**  
Yesterday

**ARN**  
[arn:aws:iam::279473426275:role/S3BucketCleanupRole](#)

**Maximum session duration**  
1 hour

[Edit](#)

[Permissions](#) [Trust relationships](#) [Tags](#) [Last Accessed](#) [Revoke sessions](#)

#### Permissions policies (2) [Info](#)

You can attach up to 10 managed policies.

[Simulate](#) [Remove](#) [Add permissions](#)

Filter by Type All types

| <input type="checkbox"/> | Policy name <a href="#">i</a>               | Type            | Attached entities |
|--------------------------|---------------------------------------------|-----------------|-------------------|
| <input type="checkbox"/> | <a href="#">AwsCostService</a>              | Customer inline | 0                 |
| <input type="checkbox"/> | <a href="#">AWSLambdaBasicExecutionRole</a> | AWS managed     | 3                 |

► **Permissions boundary** (not set)

▼ **Generate policy based on CloudTrail events**

You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. [Learn more](#)

[Generate policy](#)

No requests to generate a policy in the past 7 days.

Add an inline policy to the **S3BucketCleanupRole** IAM role with the following JSON policy.  
**Policy Name:** [AwsCostService](#)

> [S3BucketCleanupRole](#) > [Edit policy](#) 0

### Modify permissions in [AwsCostService](#) [Info](#)

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

#### Policy editor

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": [
7         "ce:GetCostAndUsage"
8       ],
9       "Resource": "*"
10    },
11    {
12      "Effect": "Allow",
13      "Action": [
14        "sns:Publish"
15      ],
16      "Resource": "arn:aws:sns:us-east-1:279473426275:DailyAWSCostAlert"
17    }
18  ]
19 }
```

[+ Add new statement](#)

Security: 0 Errors: 0 Warnings: 0 Suggestions: 0

Visual **JSON** [Actions](#) [i](#)

[Edit statement](#) [Remove](#)

**Add actions**

Choose a service

Included  
[Cost Explorer Service](#)

Available  
[AI Operations](#)  
[AMP](#)  
[API Gateway](#)  
[API Gateway V2](#)  
[ARC Region switch](#)  
[ARC Zonal Shift](#)  
[ASC](#)  
[Access Analyzer](#)

[Add a resource](#) [Add](#)

[Add a condition - optional](#) [Add](#)

10026 of 10240 characters remaining

[Check for new access](#)

[Cancel](#) [Next](#)

### Step 3: Create Lambda Function

Create an AWS Lambda function with the following configuration:

**Function Name:** **DailyAWSCostAlert**

Runtime: Python 3.14

Execution Role: **S3BucketCleanupRole**

Functions > DailyAWSCostAlert

Successfully updated the function "DailyAWSCostAlert".

Throttle Copy ARN Actions

Export to Infrastructure Composer Download

### DailyAWSCostAlert

Function overview info

Diagram Template

+ Add trigger

+ Add destination

Function ARN  
arn:aws:lambda:us-east-1:279473426275:function:DailyAWSCostAlert

Description  
-

Last modified  
11 seconds ago

Code Test Monitor Configuration Aliases Versions

#### Configuration

- General configuration
- Triggers
- Permissions
- Destinations
- Function URL
- Environment variables
- Tags
- VPC
- RDS databases
- Monitoring and operations tools
- Concurrency and recursion detection
- Asynchronous invocation
- Code signing

#### Execution role

Role name  
[S3BucketCleanupRole](#)

#### Resource summary

To view the resources and actions that your function has permission to access, choose a service.

Amazon CloudWatch Logs  
3 actions 1 resource

By action By resource

| Resource      | Actions                                                                               |
|---------------|---------------------------------------------------------------------------------------|
| All resources | Allow: logs:CreateLogGroup<br>Allow: logs:CreateLogStream<br>Allow: logs:PutLogEvents |

Lambda obtained this information from the following policy statements:

- Managed policy AWSLambdaBasicExecutionRole, statement 0

#### Step 4: Create a Test Event

Create a test event to invoke the Lambda function with the following configuration:

- Event Name: **CleanupTest**
- Event JSON:

```
{}
```

The test event "CostTest" was successfully saved.

Notifications 0 0 3 0 0 0

DailyAWSCostAlert

ThrottleCopy ARNActions

Function overviewInfo

DiagramTemplate

DailyAWSCostAlert

Layers(0)

Add triggerAdd destination

Function ARNarn:aws:lambda:us-east-1:279473426275:function:DailyAWSCostAlert

Description-

Last modified13 minutes ago

CodeTestMonitorConfigurationAliasesVersions

Test eventInfo

DeleteCloudWatch Logs Live TailSaveTest

To invoke your function without saving an event, modify the event, then choose Test. Lambda uses the modified event to invoke your function, but does not overwrite the original event until you choose Save.

Test event action

Create new eventEdit saved event

Invocation type

SynchronousExecutes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.

AsynchronousEnqueues the Lambda function for execution and returns immediately with a request ID. Function processes independently, with results optionally sent to a configured destination like SQS, SNS, or EventBridge.

Event name

CostTest

Event JSON

Format JSON

1 {  
2 {  
3 }

DailyAWSCostAlert

The test event "CostTest" was successfully saved.

Notifications 0 0 3 0 0 0

Executing function: succeeded (logs L2)

Details

{  
 "statusCode": 200,  
 "body": "10.4779798628"  
}

Summary

Code SHA-256  
GrO08OoU8RLy/c8t9CidZWG8xUbDIDh4pPqPQ7L3oZQ=

Execution time  
12 seconds ago

Function version  
\$LATEST

Request ID  
7c34ccbc-a7ad-4ad4-a0f7-5a3f0a035cbd

Duration  
945.14 ms

Billed duration  
1472 ms

Resources configured  
128 MB

Max memory used  
95 MB

Init duration  
526.27 ms

Log output

The area below shows the last 4 KB of the execution log. [Click here](#) to view the corresponding CloudWatch log group.

START RequestId: 7c34ccbc-a7ad-4ad4-a0f7-5a3f0a035cbd Version: \$LATEST  
Current Month Cost: \$10.48  
Alert: Sent  
END RequestId: 7c34ccbc-a7ad-4ad4-a0f7-5a3f0a035cbd  
REPORT RequestId: 7c34ccbc-a7ad-4ad4-a0f7-5a3f0a035cbd Duration: 945.14 ms Billed Duration: 1472 ms Memory Size: 128 MB Max Memory Used: 95 MB Init Duration: 526.27 ms

Test eventInfo

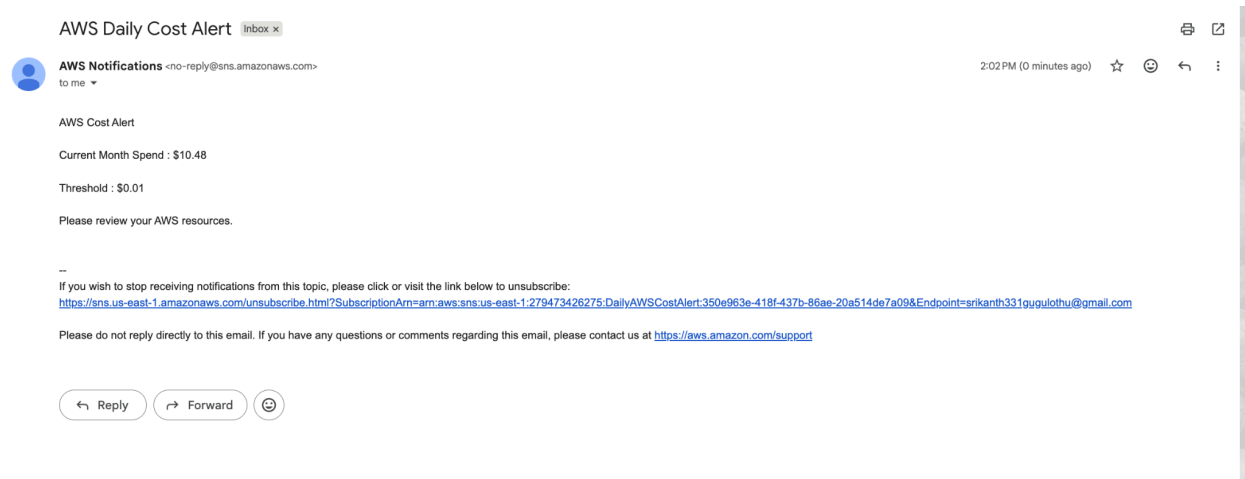
DeleteCloudWatch Logs Live TailSaveTest

To invoke your function without saving an event, modify the event, then choose Test. Lambda uses the modified event to invoke your function, but does not overwrite the original event until you choose Save.

Test event action

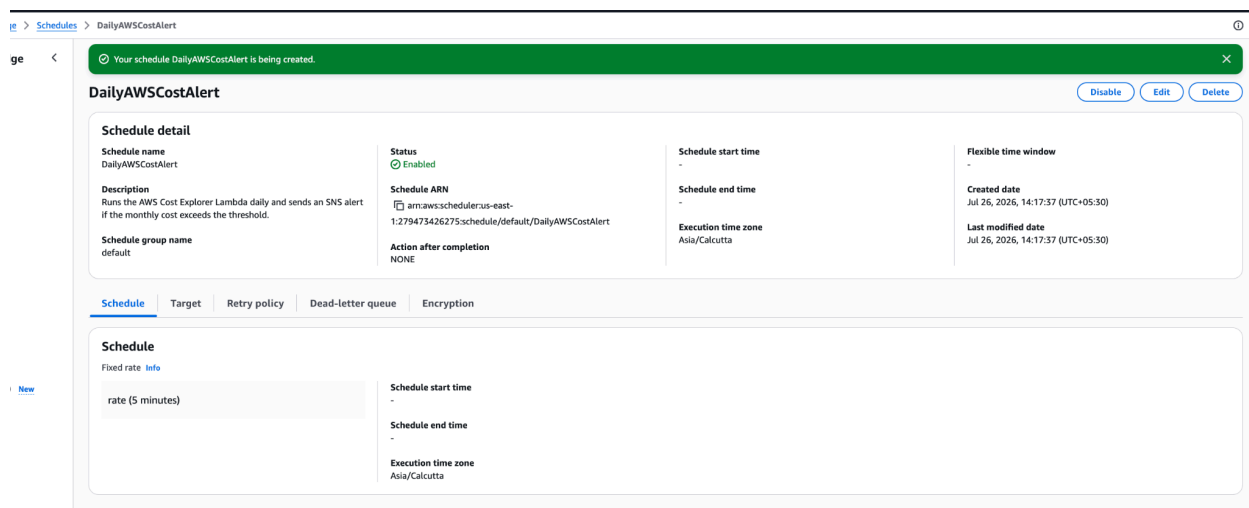
Create new eventEdit saved event

Invocation type



## Step 5: Create EventBridge Schedule

Created an EventBridge schedule named **DailyAWSCostAlert** using a rate-based schedule pattern. The target AWS service is AWS Lambda, and the selected Lambda function is **DailyAWSCostAlert**.



edules > DailyAWSCostAlert

Your schedule DailyAWSCostAlert is being created.

### DailyAWSCostAlert

[Disable](#) [Edit](#) [Delete](#)

| Schedule detail                           |                                                                                                                                 | Status                                 | Schedule start time                         | Flexible time window                                            |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|---------------------------------------------|-----------------------------------------------------------------|
| <b>Schedule name</b><br>DailyAWSCostAlert | <b>Description</b><br>Runs the AWS Cost Explorer Lambda daily and sends an SNS alert if the monthly cost exceeds the threshold. | <b>Enabled</b>                         | -                                           | -                                                               |
| <b>Schedule group name</b><br>default     | <b>Schedule ARN</b><br>arn:aws:scheduler:us-east-1:279473426275:schedule/default/DailyAWSCostAlert                              | <b>Action after completion</b><br>NONE | <b>Schedule end time</b><br>-               | <b>Created date</b><br>Jul 26, 2026, 14:17:37 (UTC+05:30)       |
|                                           |                                                                                                                                 |                                        | <b>Execution time zone</b><br>Asia/Calcutta | <b>Last modified date</b><br>Jul 26, 2026, 14:17:37 (UTC+05:30) |

Schedule [Target](#) [Retry policy](#) [Dead-letter queue](#) [Encryption](#)

### Target

[Info](#)

|                                                    |                                                                                       |                                                                                         |
|----------------------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Target</b><br><a href="#">DailyAWSCostAlert</a> | <b>Target ARN</b><br>arn:aws:lambda:us-east-1:279473426275:function:DailyAWSCostAlert | <b>Execution role</b><br><a href="#">Amazon_EventBridge_Scheduler_LAMBDA_75c8335ed7</a> |
| <b>Service</b><br>AWS Lambda                       | <b>API</b><br>Invoke                                                                  |                                                                                         |
| <b>Payload</b><br>-                                |                                                                                       |                                                                                         |

## Step 6: Cloudwatch Logs

Click on the Lambda function, then click *Test* to execute it. After the execution completes, monitor the function logs in Amazon CloudWatch Logs to verify that the Lambda function ran successfully.

management > /aws/lambda/DailyAWSCostAlert > 2026/07/26/[SLATEST]11a6f59b99034571b2843356fb6d3a

### Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns](#)

Filter events - press enter to search

Clear 1m 30m 1h 12h Custom UTC timezone Display

| Timestamp                | Message                                                                                                                                                                 |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | No older events at this moment. <a href="#">Retry</a>                                                                                                                   |
| 2026-07-26T08:48:17.788Z | INIT_START Runtime Version: python:3.14.v50 Runtime Version ARN: arn:aws:lambda:us-east-1::runtime:cb1968fe2b02c2367290ef4fb2720dc1ee0568820fe174fe63df95631f7b85       |
| 2026-07-26T08:48:18.281Z | START RequestId: 5f6e65c9-a98f-4990-9405-de9600128c7d Version: SLATEST                                                                                                  |
| 2026-07-26T08:48:18.672Z | Current Month Cost: \$10.48                                                                                                                                             |
| 2026-07-26T08:48:19.133Z | Alert Sent                                                                                                                                                              |
| 2026-07-26T08:48:19.135Z | END RequestId: 5f6e65c9-a98f-4990-9405-de9600128c7d                                                                                                                     |
| 2026-07-26T08:48:19.135Z | REPORT RequestId: 5f6e65c9-a98f-4990-9405-de9600128c7d Duration: 933.54 ms Billed Duration: 1423 ms Memory Size: 128 MB Max Memory Used: 95 MB Init Duration: 489.42 ms |
| 2026-07-26T08:49:41.397Z | START RequestId: 81948264-eb6c-45ca-a7d6-fb08439bee6c Version: SLATEST                                                                                                  |
| 2026-07-26T08:49:41.782Z | Current Month Cost: \$10.48                                                                                                                                             |
| 2026-07-26T08:49:42.009Z | Alert Sent                                                                                                                                                              |
| 2026-07-26T08:49:42.024Z | END RequestId: 81948264-eb6c-45ca-a7d6-fb08439bee6c                                                                                                                     |
| 2026-07-26T08:49:42.024Z | REPORT RequestId: 81948264-eb6c-45ca-a7d6-fb08439bee6c Duration: 624.99 ms Billed Duration: 625 ms Memory Size: 128 MB Max Memory Used: 95 MB                           |
|                          | No newer events at this moment. <a href="#">Auto retry paused. Resume</a>                                                                                               |

### AWS Daily Cost Alert

[Inbox](#)

**AWS Notifications** <no-reply@sns.amazonaws.com> to me

2:19 PM (11 minutes ago)

**AWS Cost Alert**

Current Month Spend : \$10.48

Threshold : \$0.01

Please review your AWS resources.

---

If you wish to stop receiving notifications from this topic, please click or visit the link below to unsubscribe:  
<https://sns.us-east-1.amazonaws.com/unsubscribe.html?SubscriptionArn=arn:aws:sns:us-east-1:279473426275:DailyAWSCostAlert:350e963e-418f-437b-86ae-20a514de7a09&Endpoint=srikanth331gugulothu@gmail.com>

Please do not reply directly to this email. If you have any questions or comments regarding this email, please contact us at <https://aws.amazon.com/support>